

4 · Volatility Regimes

Harry Chen

Contents

4.1	Why MACD breaks when the tape gets loud	2
4.2	What the option market already knows	3
4.2.1	Volatility cannot be computed from the price	3
4.2.2	How an option hands you a volatility	4
4.2.3	VIX	5
4.2.4	One index for equities, another for rates	6
4.2.5	From a level to a label	7
4.3	Using the label	8
4.3.1	Conditioning — the two-way sort	8
4.3.2	Residualizing — is any of it new?	9
4.3.3	Where the answer is spent	10
	Background	11
	Appendix · Notation	11

Three steps, in order. **MACD stops working when the tape gets loud** (§4.1). **The option market prices that loudness before any trailing estimate can measure it** (§4.2). The resulting label is used by conditioning on it, not by adding it to the signal (§4.3).

4.1 Why MACD breaks when the tape gets loud

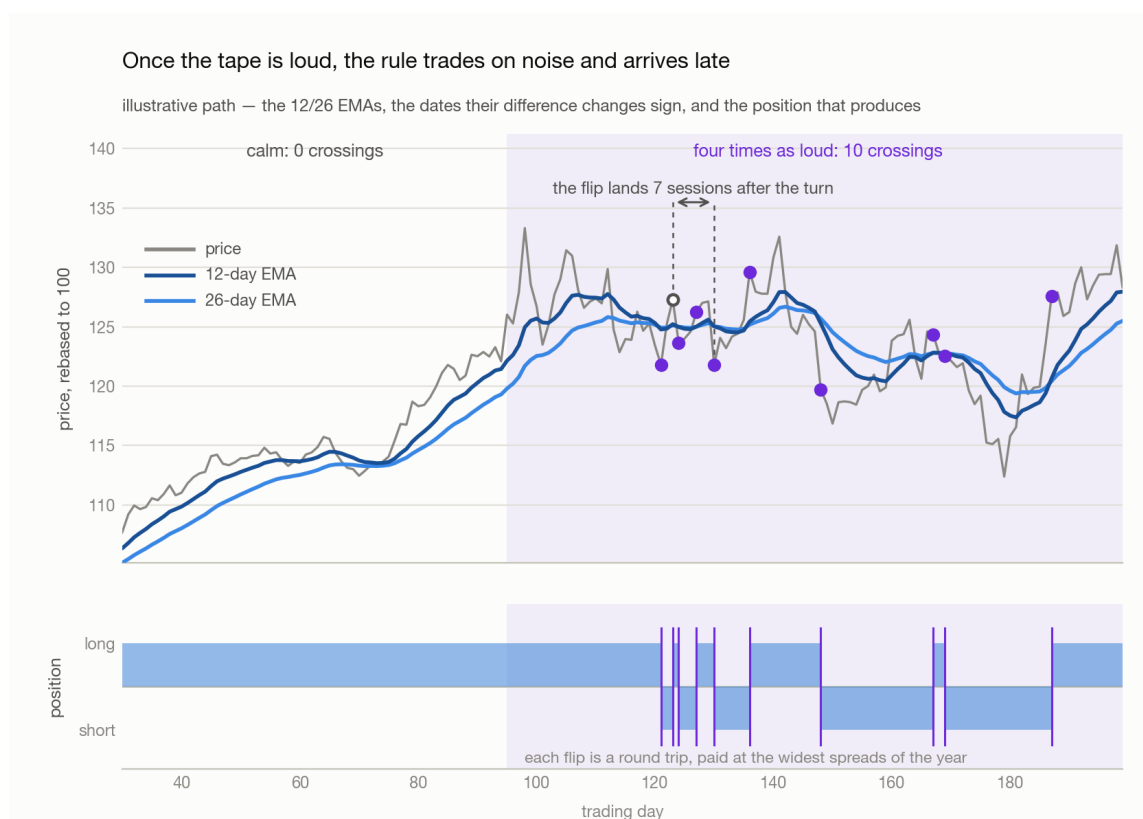


Figure 4.1: Above, one price path with a 12-day and a 26-day EMA over it. Through the calm first stretch the fast average sits clear above the slow one and the two never cross; through the shaded stretch, which has no net trend and four times the daily amplitude, they stay tangled and their difference changes sign ten times. Below, the position the rule holds: one unbroken long, then a shredded alternation of long and short blocks. Each flip is a round trip, paid at the widest spreads of the year. Illustrative.

One path, one rule, two regimes. In the calm stretch the fast average pulls clear of the slow one and stays there: no crossings, one position, held. In the shaded stretch the trend is gone and the swings are four times wider, so the two legs stay tangled and noise carries one across the other again and again.

Every flip also lands after the turn it is chasing, since an average only confirms a move that has already happened—so the rule goes short at the bottom of the spike and long at the top of the snap-back. Whipsaw is not scatter around the right answer; it is the opposite of it, repeated.

Standardizing the signal (§2.4) rescales it without moving a crossing date, and §3.4's smoother and deadband damp every date alike because neither knows the tape is violent. Clearing that ceiling means promoting the state of the market to a **variable**.

4.2 What the option market already knows

4.2.1 Volatility cannot be computed from the price

There is no way to read today's volatility off the price series. Every estimate built from past returns—a rolling standard deviation over a month, an EWMA, any weighting you like—is an average of what has already happened, so it only climbs as the new regime's returns accumulate inside its window. Shortening the window trades the lag for noise and buys nothing.

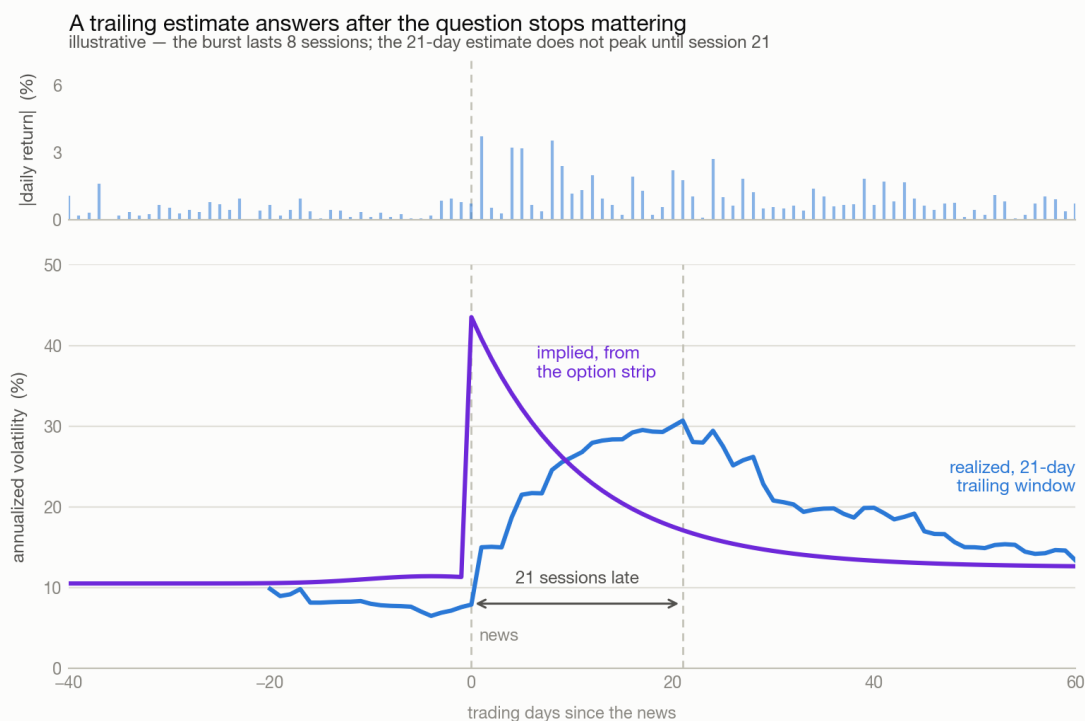


Figure 4.2: Above, the absolute daily return around a news event: quiet either side, with a burst of three and four percent moves in the sessions after it prints. Below, two answers to that event in annualized volatility. The implied line jumps on the day the news prints and decays smoothly back; the trailing 21-day line barely moves on the day and does not peak until session twenty-one, by which time the burst has been over for a fortnight. Illustrative.

The upper panel is the event and the lower panel is two answers to it. The trailing line does not peak until three weeks after the burst it is describing. **The other line is not an estimate at all**—it is a price, quoted by people who have to take a view on the next thirty days, and it moves the session the news does. Where it comes from is §4.2.2.

4.2.2 How an option hands you a volatility

An option fixes a price today for a trade that may happen later, and **the buyer chooses whether it happens**. Two of them, defined by which side that choice sits on:

	Call	Put
The holder may	buy the underlying at K	sell the underlying at K
Worth exercising when	$S_T > K$	$S_T < K$
Payoff at expiry	$\max(S_T - K, 0)$	$\max(K - S_T, 0)$
Pays off when the underlying	rises	falls
The most the buyer can lose	the premium	the premium
The most the buyer can make	unbounded	K , less the premium
Bought by someone who	wants upside without own- ing it	wants a floor under some- thing they own

with K the **strike**—the price fixed in the contract—and S_T the underlying’s price at expiry.

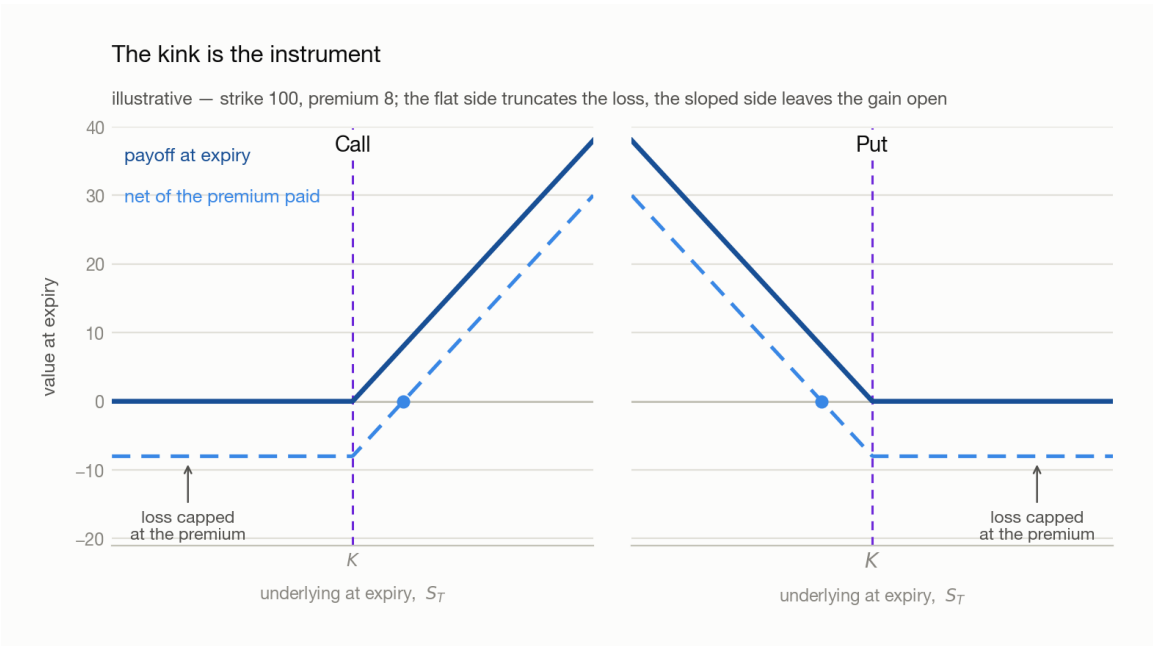


Figure 4.3: Value at expiry against the underlying’s price at expiry, with the strike K marked in both panels. Left, a call: flat at zero below the strike, rising one-for-one above it, and a dashed line one premium lower for the same shape net of what was paid. Right, a put: the mirror image. Both dashed lines are flat on their far side—the loss is capped at the premium. Illustrative.

Both payoffs are **kinked**: flat on one side of the strike, one-for-one on the other. That

kink is the whole instrument—it truncates the loss and leaves the gain open—and it is why an option is worth more when the underlying might travel further. Under Black–Scholes a call is worth

$$C = S\Phi(d_1) - Ke^{-R\tau}\Phi(d_2), \quad d_1 = \frac{\ln(S/K) + (R + \sigma^2/2)\tau}{\sigma\tau^{1/2}}, \quad d_2 = d_1 - \sigma\tau^{1/2}$$

where S is the spot price, K the strike, τ the time to expiry in years, R the risk-free rate, Φ the standard normal cumulative distribution function, and σ the volatility. Everything but σ is observable, which is what makes the equation worth inverting.

Definition 4.1 (Implied volatility). The value of σ that makes the model reproduce the option’s traded price—a forecast for the life of the contract, read out of a price rather than estimated from history.

Claim 4.2. The inversion is well defined: one price, one implied volatility.

Proof. The kink makes both payoffs convex in S_T . Raising σ spreads the distribution of S_T while leaving its forward mean where it was, and the expectation of a convex function rises under a mean-preserving spread. The price is therefore strictly monotone in σ , and a monotone map is invertible. $\square$

There is no closed form for the inverse, so σ is solved numerically—bisection needs about twenty steps to reach four decimals, and monotonicity is what guarantees it converges. At the money the whole thing collapses to something you can do in your head:

$$C \approx 0.4 S \sigma \tau^{1/2} \quad \implies \quad \sigma \approx \frac{C}{0.4 S \tau^{1/2}}$$

Example 4.3. A 30-day at-the-money call trades at 2 percent of spot. Quote the price as a fraction of spot and S cancels, leaving a denominator of $0.4\tau^{1/2} \approx 0.115$ at $\tau = 30/365$, so $\sigma \approx 0.02/0.115 \approx 0.17$ —the market is charging for a **17 percent annualized** move over the next month. Read the same contract a day later at 3 percent of spot and the forecast is 26 percent: the number moves with the quote, which is the entire point.

Note (One contract is not the market). That 17 percent is the volatility charged **at that strike**. Quotes on other strikes imply different numbers—the smile—so a single contract’s answer is a property of one contract. §4.2.3 reads the whole strip at once to get rid of that dependence.

4.2.3 VIX

Definition 4.4 (VIX). A constant-30-day implied volatility for the S&P 500, built from the whole strip of out-of-the-money options rather than from any one of them. For a single expiry the strip gives the implied **variance** directly, with no model to invert:

$$\sigma_\tau^2 = \frac{2e^{R\tau}}{\tau} \sum_i \frac{\Delta K_i}{K_i^2} Q(K_i) - \frac{1}{\tau} \left(\frac{F}{K_0} - 1 \right)^2$$

where $Q(K_i)$ is the mid price of the out-of-the-money option struck at K_i , ΔK_i the gap between neighbouring strikes, F the forward level implied by put-call parity, and K_0 the first strike below F . Every term is a traded price, so the sum is the market's own variance rather than one contract's.

Two steps turn that into the published number. **Take the two expiries with τ between 23 and 37 days**—listed contracts do not fall on a 30-day schedule, and the band is what guarantees one on each side of the target—and **interpolate their variances linearly in τ** onto exactly 30 days. Then annualize, take the root, and quote it in percentage points:

$$\text{VIX}_t = 100 \left(\sigma_{30,t}^2 \right)^{1/2}$$

Note (Decide whether your formula wants the variance or the volatility). The published level is already a volatility—a VIX of 20 means an annualized 20 percent—because the strip computes a variance and the last step takes its root. Variance is what is additive, across time and across independent sources of risk; volatility is not. Anything that rescales a horizon or adds contributions needs $(\text{VIX}_t/100)^2$; anything compared against a return in the same units needs the level. The error is invisible when made: a missing square root still leaves a plausible positive number.

Note (What the forecast is worth is a separate question). Implied volatility is what the market charges, not what will happen—insurance is sold above its expected cost, and that gap is a strategy in its own right. Here the index is only a **state label**, for which a live consensus is enough.

4.2.4 One index for equities, another for rates

Claim 4.5. One equity index's implied volatility labels most sleeves, even though it is computed from one market.

The reason is what a panic does to correlations. In calm markets the sleeves move mostly apart; in a large enough shock the same levered participants sell all of them at once to raise cash and the cross-asset correlations run toward one. The states in which the label matters most are exactly the states in which one market's fear is every market's fear—an empirical regularity rather than a theorem, and better at the extremes than in the middle.

Sleeve	Regime index	What its spikes are about
Equities, credit, most commodities	VIX	growth and earnings shocks, and forced deleveraging
Rates, and anything priced off the curve	MOVE	policy surprises, inflation prints, auction stress

Note (Rates are the genuine exception). Treasury volatility is close to unrelated to equity volatility, because the two are frightened by different things. A rate sleeve labelled by VIX is labelled wrongly, and **MOVE** is the same construction run on Treasury options. Neither index is in CTA_data/ (100 · The Dataset), so both have to be fetched before §4.3 can be run.

4.2.5 From a level to a label

Definition 4.6 (Volatility regime). A stretch of dates over which the amplitude of returns is roughly constant and materially different from the stretch on either side; a **regime shift** is the boundary between two of them.

§4.3 needs a small set of such states, named, because a continuous variable cannot be a row of a table. Write v_t for the index level on date t ; its distribution is the obstacle.

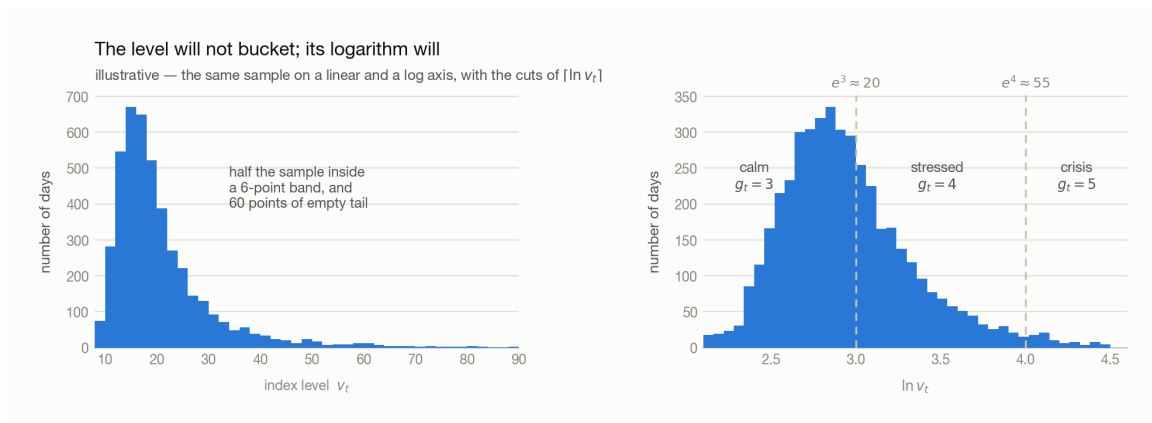


Figure 4.4: The same illustrative sample of volatility index levels, twice. Left, against the raw level: the mass piles into a narrow band and trails off into a long near-empty tail, half the sample sitting inside six points. Right, against the natural logarithm of the same values: close to symmetric, and the two integer cuts at 3 and 4—about 20 and about 55 on the level—divide it into calm, stressed and crisis.

The level spends most of its life in a band a few points wide and occasionally reaches four times that, so equal-width bins put nearly everything in the first one—the opposite of what §2.3.2.2’s bucket test needs.

Definition 4.7 (Log-ceiling bucket). Take the natural log of the level and round up to the next integer, $g_t = \lceil \ln v_t \rceil$, where g_t is the regime label on that date. Because $\ln$ compresses the sparse upper tail and stretches the crowded lower band, the integer cuts land at multiplicatively spaced levels:

Label g_t	Level range	Reads as
3	$v_t \in (7.4, 20.1]$	calm—the ordinary tape
4	$v_t \in (20.1, 54.6]$	stressed
5	$v_t \in (54.6, 148.4]$	crisis

The boundaries are e^3 and e^4 , close to where a practitioner would have drawn them by hand. That is the construction’s virtue: **the cuts use no data, so a date’s label means the same thing in 2015 and in 2020**. Writing $g_t = \lceil c \ln v_t \rceil$ puts the boundaries at $e^{k/c}$ and recovers as many buckets as wanted; c is a hyperparameter like any other, and choosing it by which value produces the best-looking grid is what [5 · Modelling](#) is about.

The alternative is [§2.3.2.1](#)’s own machinery applied to the index: rank the dates by v_t and cut at the quintiles.

	Log-ceiling	Rolling quantile
Boundaries	fixed, the same in every year	move with the window
Group sizes	very unequal—crisis may be 3 percent of the sample	equal by construction
A label means	an absolute level of market fear	a rank against the recent past
Comparable across time	yes	no—the same v_t can be G5 one year and G3 the next
Look-ahead risk	none, the cuts use no data	real, if the window is not trailing

Note (The ranking must be rolling, never full-sample). A quintile computed over the whole history uses 2020’s levels to label 2015’s dates, which is look-ahead bias of the plainest kind ([§2.6](#))—and it is easy to commit here, because the index feels like context rather than like data.

4.3 Using the label

Each date now carries three numbers: a signal value, a forward return, and a regime label g_t . Two different questions can be asked of that—whether the edge **differs** by regime ([§4.3.1](#)), and whether the regime is **new information** the signals do not already carry ([§4.3.2](#)).

4.3.1 Conditioning — the two-way sort

Put the signal buckets across and the regime labels up, and report the mean forward return in every cell. It is [§2.3.2](#)’s bar plot run once per regime—equivalently a **conditional model**: sort on the state first, then on the signal within it.

Read it by rows, and the reading is the finding. A staircase in the calm row that flattens through the stressed row and disappears in the crisis row is [§4.1](#)’s picture measured on your own data: same rule, same parameters, less and less to show for them. The one-dimensional plot of [2](#) is the sample-weighted average of the three rows—it reports the calm row lightly diluted and never mentions that the other two exist.

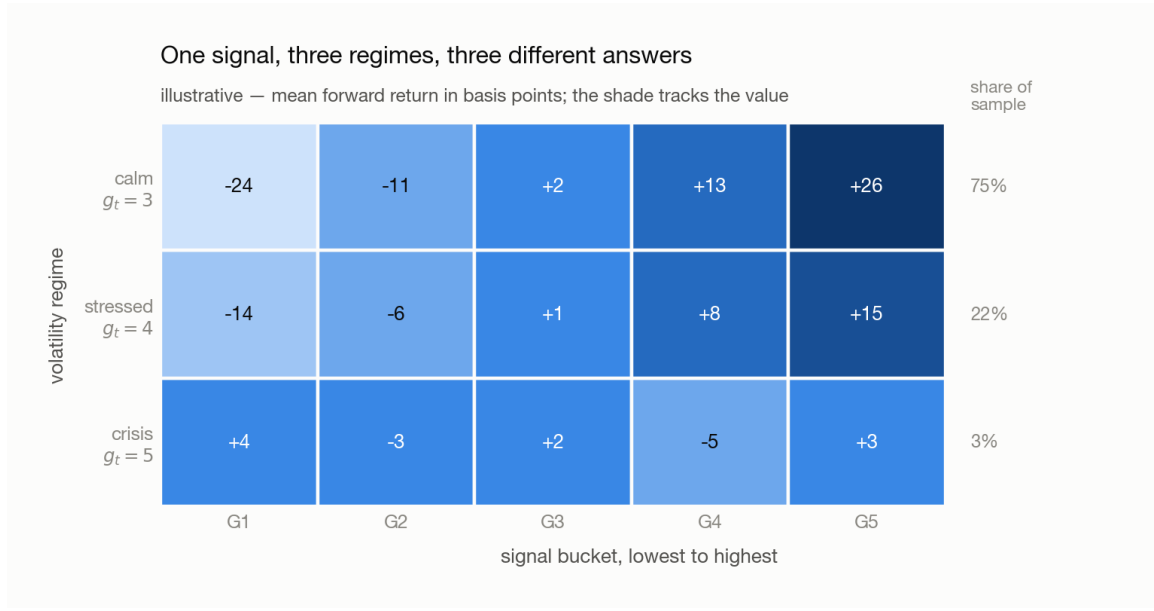


Figure 4.5: Mean forward return in basis points, signal buckets G1 to G5 across and volatility regime up: calm, stressed, crisis. The calm row, holding three-quarters of the sample, rises monotonically; the stressed row rises about half as steeply; the crisis row, 3 percent of the sample, has no ordering at all and its shading is flat across the row. Illustrative.

Note (Keep the regime axis coarse, and print the counts). The cells partition the same sample, so a one-way bar’s $N/5$ dates become $N/15$ across three rows and every error bar (§2.3.2.3) widens by $3^{1/2} \approx 1.7$ —at five rows, $5^{1/2} \approx 2.2$. Worse, the row you most want is the thinnest: crisis cells hold roughly $N/167$ dates, error bars about 5.8 times wider, so **a flat crisis row is equally consistent with an edge nobody has enough crisis days to measure**. Report the count beside every mean, and treat a row you cannot measure as unknown rather than as zero.

4.3.2 Residualizing — is any of it new?

Regress the candidate on the incumbents and keep what is left:

$$x_t = \beta_0 + \sum_j \beta_j x_{j,t} + \epsilon_t$$

where x_t is the candidate signal on date t , $x_{j,t}$ the signals already in the book — momentum and MACD, here— β_j their fitted coefficients, and ϵ_t the residual. Then run §2.3.2’s bar plot on ϵ_t in place of x_t .

Claim 4.8. If the staircase survives on ϵ_t , the candidate carries an edge the existing signals do not already own.

Proof. Least squares sets the derivative of the summed squared error to zero, giving one normal equation per regressor— $\sum_t \epsilon_t x_{j,t} = 0$ for every j , and $\sum_t \epsilon_t = 0$ from the

intercept. Zero sum of products on a centred variable is zero sample covariance, so $\rho(\epsilon, x_j) = 0$ for every incumbent. By §2.1.2 a portfolio’s expected return runs entirely through the correlation between what sets the weights and the forward return, so weights built on ϵ_t earn whatever they earn somewhere the book is not standing. $\square$

Note (Orthogonal is not independent). Zero *linear* correlation is all least squares buys. A residual that is large exactly when MACD is large **in absolute value** is a function of MACD the regression cannot see—and that is not a corner case here, since the candidate is a variable about σ and $|\text{MACD}|$ widens with σ too. Add the transform you suspect as an extra regressor and residualize against that too.

Note (With fifty signals, regress against the model’s output). Fitting a candidate against fifty regressors on overlapping daily data fits noise long before it exhausts the information, and fifty pairwise regressions give answers that do not compose. The book already produces one number per asset per date—its combined prediction p_t . Use that single series as the regressor: one residual, and the question it answers is the one actually being asked.

4.3.3 Where the answer is spent

Often not in the return model at all. The natural home for §4.3.1’s grid is the **optimizer**—a penalty that rises with the regime, an exposure cap that binds in the crisis row, a volatility target that scales the whole book—which is why the grid earns its place even when §4.3.2’s residual comes back flat. Answering *yes* to §4.3.1 and *no* to §4.3.2 is a common and perfectly good result, and performance evaluation is where it is spent.

Add one dimension per experiment, and record what it bought—what the step adds alone, whether it overlaps what is already there (§4.3.2’s residual), and which direction it is useful in, since a variable can predict return poorly and volatility well. The record has to pass interpretability in an operational sense: move an input and you should be able to say in advance, roughly, what the output does. Measuring 8 bp where you expected 10 is fine; *“it moved and I do not know by how much”* is not, and it is what adding several things at once guarantees.

Note (The scorecard is regime-dependent too). Every chart above reports mean forward return with an error bar, for the reasons §2.5 sets out. A Sharpe ratio has the tape’s volatility in its own denominator, so the same number means different things in different rows: 0.8 earned in a year when the index itself ran at 0.4 is a far harder result than 0.9 earned in a year when buying the index and doing nothing paid 0.8. Collapsing the regime axis into one ratio destroys exactly the context needed to read the ratio.

Background

Why is an option the instrument a volatility forecast comes from?

- **Its value depends on the range, not the direction.** The flat half of the payoff truncates the losses, so a wider distribution of outcomes is worth strictly more—which makes an option the most volatility-sensitive thing quoted, and §4.2.2’s inversion stable enough to be worth doing.
- **Out-of-the-money contracts are close to pure volatility bets.** One struck far above the price pays only if the underlying travels an unusual distance, so its price is almost entirely a statement about the tails—which is why VIX reads the whole strip, not the at-the-money contract.
- **The forecast has a fixed horizon attached.** Every contract has an expiry, so an implied volatility is always *volatility over the next τ days*—which is what allows two of them to be interpolated onto a constant 30-day horizon.

Appendix · Notation

Throughout, t is the date. The regime index is a property of the market rather than of one asset, so it carries no asset subscript s .

Symbol	Means	First used
σ	the volatility of the underlying—the amplitude of its move, without regard to sign	§4.2.2
C, S, K, τ, R	a call’s price, the spot price, the strike, the time to expiry in years, and the risk-free rate	§4.2.2
Φ, d_1, d_2	the standard normal cumulative distribution function and the two arguments Black–Scholes feeds it	§4.2.2
S_T	the underlying’s price at expiry	§4.2.2
$Q(K_i), \Delta K_i, F, K_0$	one out-of-the-money option’s mid price, the gap between neighbouring strikes, the forward level, and the first strike below it	§4.2.3
$\sigma_\tau^2, \sigma_{30,t}^2$	the implied variance the strip gives for one expiry, and the same interpolated onto 30 days	§4.2.3
v_t, g_t, c	the index level on that date, the regime label cut from it, and the multiplier setting how fine the cuts are	§4.2.5
$x_t, x_{j,t}, \beta_j, \epsilon_t$	the candidate signal, the signals already in the book, their fitted coefficients, and the residual	§4.3.2

Symbol	Means	First used
p_t, N	the model's combined prediction, and the number of dates in the sample	§4.3

Note (Collisions to watch). σ is a market-wide volatility here where §2.4's $\sigma_{s,t}$ is one asset's trailing estimate, and R is the risk-free rate discounting an option where §2.5's r_f is the same rate subtracted from a portfolio return. Uppercase S is a spot price and lowercase s is 2's asset index; Φ is a normal CDF and N a sample size, which is why the CDF is written Φ rather than N . Lowercase g_t is a regime label on the vertical axis while §2.3.2's uppercase G1 ... G5 are signal buckets on the horizontal one, and §4.3.1's grid has both at once. ϵ_t here is a signal residual after regressing on other **signals**; §2.2's ϵ is the part of the forward **return** no signal reaches.